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A recurrence plot-based approach for Parkinson's disease identification

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HIGHLIGHTS

- We proposed the usage of recurrence plots for Parkinson's disease identification.
- Recurrence plots provided more discriminative representations.
- Significant improvement was accomplished using the proposed approach.
- An average accuracy of over 87% was achieved.

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ABSTRACT

Parkinson's disease (PD) is a neurodegenerative disease that affects millions of people worldwide, causing mental and mainly motor dysfunctions. The negative impact on the patient's daily routine has moved the science in search of new techniques that can reduce its negative effects and also identify the disease in individuals. One of the main motor characteristics of PD is the hand tremor faced by patients, which turns out to be a crucial information to be used towards a computer-aided diagnosis. In this context, we make use of handwriting dynamics data acquired from individuals when submitted to some tasks that measure abilities related to writing skills. This work proposes the application of recurrence plots to map the signals onto the image domain, which are further used to feed a Convolutional Neural Network for learning proper information that can help the automatic identification of PD. The proposed approach was assessed in a public dataset under several scenarios that comprise different combinations of deep-based architectures, image resolutions, and training set sizes. Experimental results showed significant accuracy improvement compared to our previous work with an average accuracy of over 87%. Moreover, it was observed an improvement in accuracy concerning the classification of patients (i.e., mean recognition rates above to 90%). The promising results showed the potential of the proposed approach towards the automatic identification of Parkinson's disease.

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1. Introduction

First described by James Parkinson in 1817, Parkinson's disease (PD) is a neurodegenerative disease affecting around 7 to 10 million people worldwide, being nearly 1 million people only in the United States of America. Also, according to the National Parkinson's Foundation [1], the number of new cases diagnosed each year ranges from 50,000 to 60,000 individuals. PD is caused by the loss of a neurotransmitter called *Dopamine* [2], being a chronic, progressive and multi-lesion disease. The lack of Dopamine increases the malfunctioning of the brain affecting the communication among its cells. As a consequence, messages are not sent correctly, which may lead to cases of depression, sleep disturbances, dementia, depression, psychotic features, and autonomic dysfunction [3]. PD patients may also face motor dysfunctions, such as tremor, and an increase in the risk of falls and mobility disabilities due to alterations in gait and posture. It has a tremendous negative impact on the daily routine of PD patients and their families, thus reducing their quality of life [4–6].

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Although non-lethal, life expectancy is shorter for PD patients compared with the general population. Its diagnose is usually performed using a clinical exam and by a neurologist with expertise in movement analysis. The current treatments for PD symptoms include therapy and drugs known as dopaminergic medications, being the *Levodopa* (L-dopa) the most widely used for such purpose. Another widely employed treatment is the Deep Brain Stimulation, which is a surgical procedure that delivers electrical pulses to brain cells to reduce the effects of the symptoms.

The science does not measure efforts to improve the quality of life regarding PD patients [7–10]. Computer-aided detection and diagnosis systems based on image processing, neural networks, and other techniques have been widely applied in the pursuit of better results in both treatment and aid of diagnosis [11]. One of the motivations for the development of such tools comes from the fact that pattern recognition and other techniques may be able to detect subtle signs assigned to PD that might not be noticed by a human. Also, they may provide some relevant information to support the final diagnosis. Spadotto et al. [12], for instance, proposed the automatic PD identification using the Optimum-Path Forest (OPF) [13,14] classifier. A later work of the same group proposed the selection of a subset of features through evolutionary-based techniques, which helped to improve PD identification accuracy [15–17].

Most works that address automatic PD identification cope with voice-based data. Procedures to identify voiced and unvoiced (silent) periods have been actively pursued to analyze continuous speech samples since most techniques that quantify periodicity and regularity in voice signals are applied in the voiced regions only [18]. Das [19] compared different classification methods (e.g., neural networks, decision trees, and regression) for the diagnosis of PD. The experiments were carried out in a dataset comprised of biomedical voice measurements collected from 31 people, being 23 diagnosed with PD and eight healthy individuals. The approach was evaluated under several methods, with the best results achieved by neural networks with an accuracy of around 92.9%.

However, the clinical diagnosis of PD can also be performed by the identification of some motor signs such as bradykinesia, tremor, rigidity, and even postural instability [3]. A few studies were dedicated to monitor, assist, and aid the detection of PD based on signals collected from accelerometers and other sensors placed in wearable devices and biometric pens, which can provide valuable information of any existing motor dysfunction. Recently, Mamun et al. [20] proposed a cloud-based framework to allow the diagnosis and monitoring of PD. The individual interacts with the doctor through mobile devices connected to the internet. Voice samples are recorded and have features extracted for the diagnosis using machine learning techniques.

Khan et al. [21] worked with different accelerometer data including those from PD patients performing daily activities to find an optimum sampling for human activity recognition. Sama et al. [22] proposed to detect the freezing of gait (FoG), another common symptom in PD patients, based on data collected by a single-waist accelerometer in home environments. The work aimed at the monitoring of FoG to apply real-time cueing strategies by selecting a subset of optimal features. Bächlin et al. [23] introduced a wearable assistant for PD patients that present FoG called "wearable health assistant", which aims to reduce the number and length of their motor blocks, and thus increase their safety while walking by performing an online movement analysis and providing a rhythmic auditory signal that stimulates the patient to resume walking.

A work on accelerometer-based posture analysis in PD was proposed by Palmerini et al. [24]. A total of 175 measurements are computed from the accelerometer data, and a feature selection technique was applied to select the most discriminative set of measurements to characterize PD and healthy control (HC) subjects. The subset of measurements was used to quantify tremor, acceleration, and displacement of body sway.

Moreover, a significant number of works explored tremor, which is the most common disorder in PD patients. Rigas et al. [25] proposed an automated method to assess both resting and action tremors by estimating the type and severity of data acquired from accelerometers attached to specific positions at a patient's body. The estimation is based on features extracted from the signals and Hidden Markov Models. Abdulhay et al. [26] used tremor along with gait and machine learning techniques for the diagnosis of PD. They extracted features from signals recorded by sensors placed underneath the patients' feet and at the forefinger during deep brain stimulation.

Pereira et al. [27,28] proposed the identification of PD using visual features extracted from a handwriting exam comprised by tasks supposed to be nontrivial for PD patients. The patient is asked to perform some drawings (spirals and meanders) over a guideline that are compared against templates, whose difference between them (tremor) provides information to feed machine learning techniques to classify an individual as either healthy or PD patient. In a later work, Pereira et al. [29,30] drove their approach to a deep learning application focusing on the signals captured during the same handwriting exams from [27,28] using a smart pen.[1] The signals were mapped into the image domain with different resolutions using visual rhythms and used as input to a Convolutional Neural Network (CNN). Despite the possible loss of information caused by the mapping of the signals, the approach was able to outperform those proposed in [27,28].

Although the recent advances achieved by computer-aided detection and diagnosis systems, there is still room to improve research in automatic PD identification. Moreover, any improvement is relevant to early diagnosis. As shown in the work of Pereira et al. [29], signals turned to be a promising data source for the PD identification since more subtle variations can be detected. In this work, we propose an extension of the work performed by Pereira et al. [29] by applying the *recurrence plot* technique (RP) to map signals into the image domain to further feed a CNN. The RP is a technique proposed by Eckmann et al. [31] that enables to visualize repeated events of higher dimension through projections onto a two or three-dimensional representation. Moreover, CNNs are capable of learning very discriminative features, and their application has provided many promising results as well [32].

In short, the contributions of this work are: (i) to the best of our knowledge, this is the first application of recurrence plots for modeling signals in the context of PD identification, (ii) the investigation of the proposed approach performance under different configurations, and (iii) the accuracy improvement provided by the usage of RP with respect to our previous work [29].

The remainder of this work is organized as follows. The proposed approach, as well as material and methods used as evaluation tools, are described in Section 2. The experiments are detailed in Section 3, and Section 4 states conclusions and future works.

2. Proposed approach and methodology

This section describes the material and methods employed to evaluate the proposed approach for automatic PD patients identification. At a glance, signals were recorded when patients and healthy control (HC) individuals perform some specific drawings. The following step models the signals into images via the recurrence plot method whose output highlights the patterns of each class. Further, a machine learning algorithm performs the

[1] https://www.oth-regensburg.de/index.php?id=5312/biometrics.html.

Table 1
HandPD dataset description.

	HC	PD
Female	10	4
Male	11	10
Total	**21**	**14**

classification based on the patterns learned from the output images (i.e., either from a CNN or the raw image), as depicted in Fig. 1. The following sections provide details of each step, including the process of data acquisition used in the experiments.

2.1. HandPD dataset

The development of the HandPD dataset was motivated by the fact that datasets comprising images usually employ high-end image technology, which is expensive and may be invasive as well. Also, available signal-based datasets may not reflect the real world because of the low number of samples and might be biased [27].

The HandPD dataset comprises data collected by the Medical School of the São Paulo State University (UNESP), Botucatu, Brazil, from handwriting exams performed by potential patients in 2010. The exam is comprised of six tasks that are supposed to be nontrivial to PD patients. In the first task (exam (a) in Fig. 2), the individual is asked to draw a circle 12 times in the same place continuously. In the second task (exam (b) in Fig. 2), the individual performs the same movement as in the exam (a), but with its hand in the air. The third and fourth (exams (c) and (d) in Fig. 2, respectively) tasks stand for drawing the spirals and meanders, respectively, over the four templates, only once each, from the inner to the outer part. The last two tasks, i.e., exams (e) and (f) in Fig. 2, stand for the diadochokinesis test, which is basically hand–wrist movements performed with both hands. The activities concern the quantification of the motor activity of HC individuals and PD patients by analyzing the level of tremor when drawing spirals and meanders.

In its first version (HandPD V1), Pereira et al. [27] built the dataset[2] considering only the images of the drawings acquired from a total of 35 exams divided into two classes: (i) HC individuals and (ii) PD patients. Table 1 presents the information about HandPD dataset.

The drawings (spirals and meanders) were manually cropped from the forms and stored in the "jpg" extension (Fig. 3). Therefore, the former works using the HandPD dataset were driven by image processing techniques for the identification of PD [27,28]. Since we are dealing with information from hand tremors, it was noticed that the signals were also an important source of information since they can provide more detailed information about the tremor intensity. Moreover, a newer version of the HandPD (HandPD V2) dataset was released by Pereira et al. [29], which comprises all drawings from HandPD V1, as well as their corresponding signals in their raw format concerning all six tasks performed by the individuals. The signals were collected from sensors placed in the pen used by the individuals to perform the drawings. Fig. 4(a) and (b) depict the signals corresponding to the drawings illustrated in Fig. 3(a) and (b), respectively. It can be observed a "noisier" data in the PD patient signal over the healthy individual.

2.2. Modeling signals into images

The PD identification problem is modeled as an image classification task using RP-based images and CNNs. Each data is represented by a set of 6 arrays (i.e., signal channels) whose length stands for the acquisition time (milliseconds), and further mapped into images through the recurrence plot methodology. Due to the patients' different paces, the signals' length were limited to the same extent of the shortest among all.

Proposed by Eckmann et al. [31], the recurrence plot is a tool for non-linear analysis of time series data that allows to obtain visual information and to investigate recurrent behavior in time series. The RP provides an array of points that can be seen as a squared image of size $N \times N$, being N the number of considered states. The points of the array are placed at the coordinate (i, j) whenever $x(j)$ is the most similar state to $x(i)$, in which $x(i)$ is a state in the time series.

The original formulation considered binary or gray-scale images only. To compose an RGB image, each acquisition session needs to be re-scaled into three channels, and further normalized between 0 and 1, as proposed by Faria et al. [33]. As we are trying to distinguish the writing patterns between healthy and PD patients, we discarded both channels 1 and 4, since we observed that the microphone and the displacement in the x-axis are similar for all acquisition sessions. Finally, regarding the last channel, we employed arithmetic mean between channels 5 and 6. Fig. 5 illustrates a few RGB RP-based images considering both meanders and spirals drawn by healthy and PD patients, respectively. Distinct patterns can be observed between HC and PD patient samples.

In regard to the CNN, we used three different architectures for comparison purposes: (i) CIFAR10_quick,[3] (ii) ImageNet[4,5] and (iii) LeNet architecture.[6] The architectures are trained using mini-batches of size 16 with 10,000 iterations and default hyperparameters defined by these architectures. Transfer learning was not employed in this work since it is very much a domain-specific application.

We used the implementations provided by the well-known Caffe library[7] [34], which is developed under a General-Purpose computing on Graphics Processor Units. Figs. 6 and 7 illustrate the architectures of CIFAR10_quick and LeNet, respectively. Each type of layer is represented by a different box, being "conv" the convolution layer, "pool" the pooling layer, the rectified linear unit activation as "relu", normalization as "norm", "ip" the inner product or fully connected layer, "accuracy" for the final accuracy output, and "loss" for the output of the loss function.

2.3. Classification

The experiments aimed at classifying the RP-based images computed from the original signals of meanders and spirals drawn by the HC group and PD patients. Two different image resolutions (i.e., 64×64 and 128×128 pixels) were used to assess the robustness of CNNs. For comparison purposes, an experiment over the raw RP images served as a baseline as well. Notice that by raw RP images, it is meant the pixels values themselves are used to composing the feature vector that feeds the classifiers.

The experiments used datasets comprising meanders and spirals data, with a total of 308 images each (84 HC and 224 PD patients). The impact of the number of training samples over the classification accuracy is evaluated through two distinct scenarios, i.e., 75% and 50% for testing purposes.

2 http://wwwp.fc.unesp.br/~papa/pub/datasets/Handpd.

3 https://github.com/BVLC/caffe/blob/master/examples/cifar10/cifar10_quick_train_test.prototxt.

4 https://github.com/BVLC/caffe/blob/master/models/bvlc_reference_caffenet/train_val.prototxt.

5 Due to the network size, we did not opt not to employ a figure of its own, however, one can check the full architecture on its default prototxt file.

6 https://github.com/BVLC/caffe/blob/master/examples/mnist/lenet_train_test.prototxt.

7 http://caffe.berkeleyvision.org.

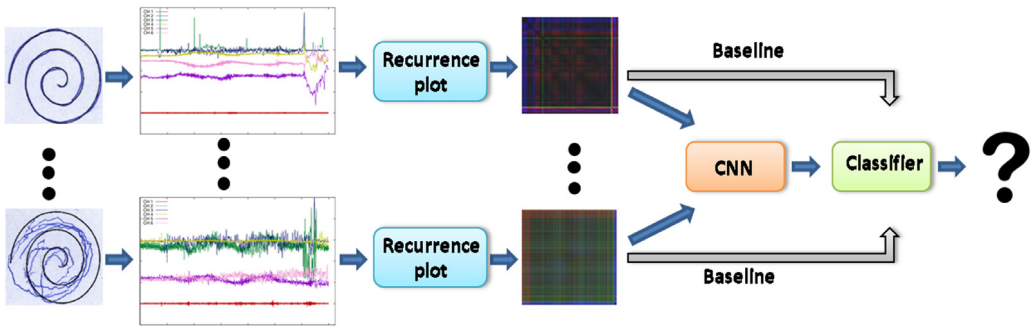

Fig. 1. Workflow of the proposed approach for computer-aided PD identification.

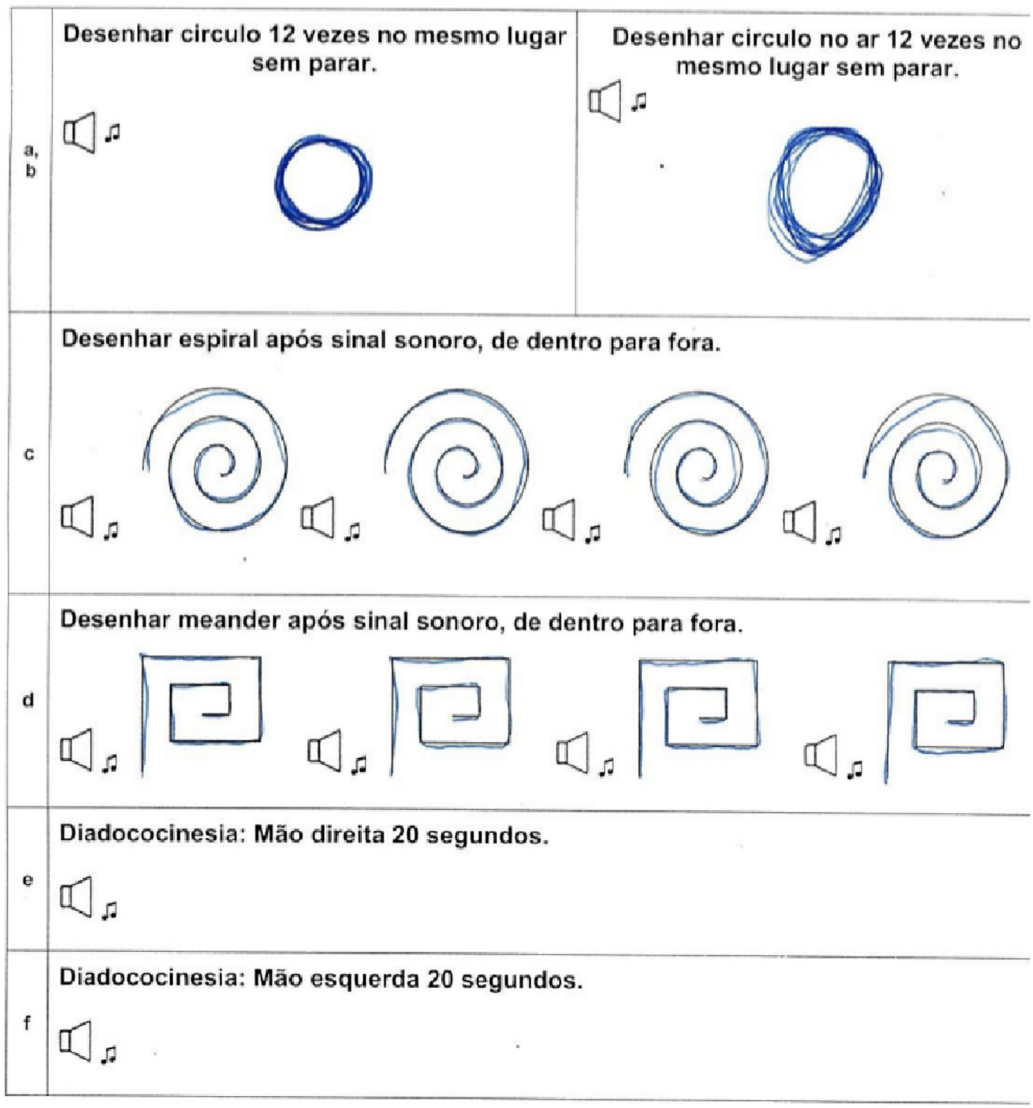

Fig. 2. Exam sample used to assess the handwritten dynamics.
Source: Adapted from [28].

Although any supervised machine learning technique could be employed, we opted to use the OPF classifier [13,14], which is parameterless and employed in our previous works. In order to have a fair comparison, notice that all parameters (i.e., CNNs' hyper-parameters) were set as in Pereira et al. [29], and those that are not mentioned in that work have their values set empirically or by a default value.

3. Experimental results

The proposed approach is evaluated using the Meander and Spiral datasets under different configurations concerning image resolution and training set size. The experimental results are organized by neural architecture and pair [image resolution, training set size] for both Meander and Spiral datasets. The accuracy rates were computed using the standard formulation, i.e., $(1 - \frac{\#\ misclassifications}{total\ \#\ samples})*$

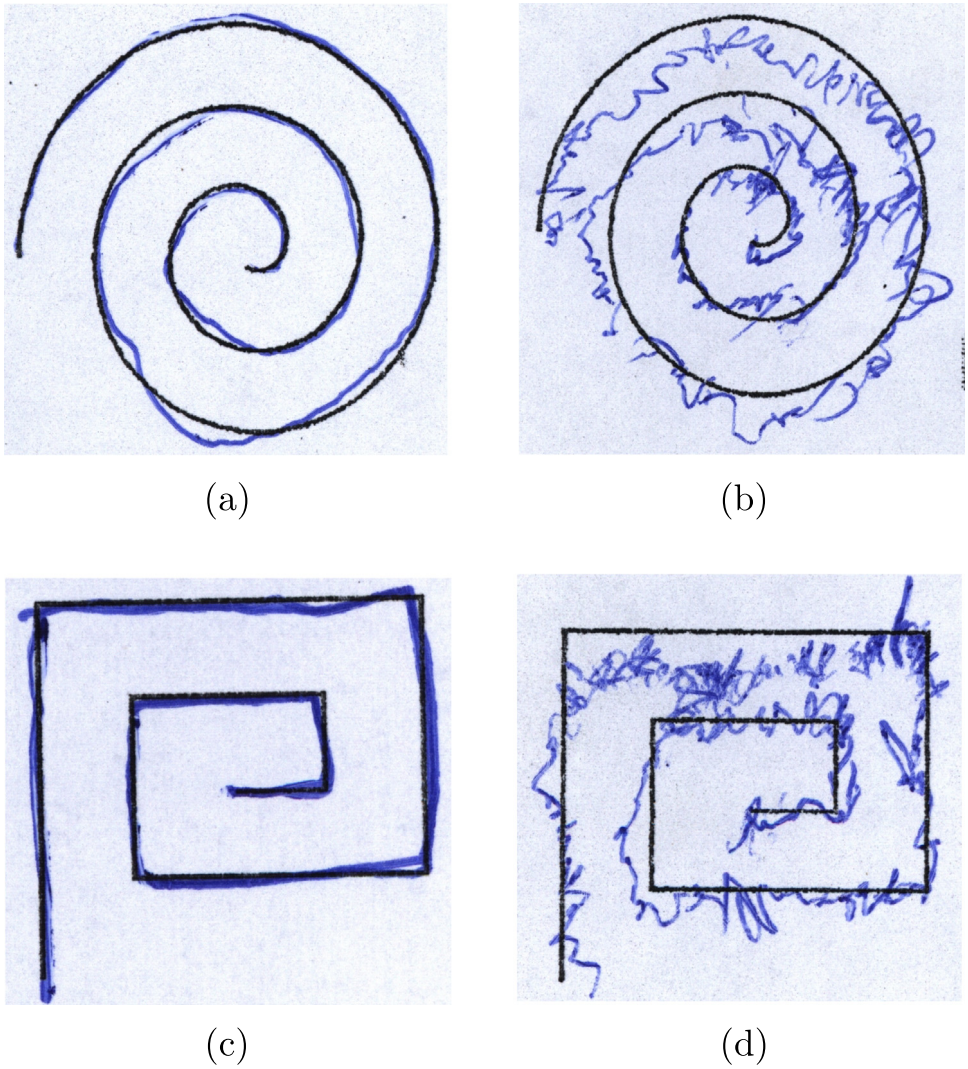

Fig. 3. Drawings samples from spirals (a) HC and (b) PD patient, as well as meanders (c) HC and (d) PD patient.

100. The best results of each pair are highlighted in bold according to the Wilcoxon signed-rank test [35] with a significance of 5% over a hold-out validation of 20 runs. Underlined results are the best ones regardless the configuration.

3.1. Meander dataset

We first discuss the results related to the overall accuracy rates concerning the Meander dataset, as presented in Table 2. One can observe that CIFAR-10 achieved an average gain of 22.94% concerning the approach proposed by Pereira et al. [29]. Meanwhile, LeNet achieved an improvement of 58.58% and 41.42% using training set sizes of 50% and 75%, respectively. As the results obtained in the previous work, ImageNet achieved the highest accuracy rates in all scenarios and outperformed itself in the configurations where the training set has 75% of the dataset samples (Table 2).

However, one can observe that RP-based representations posed a challenge to OPF that showed a decrease in improvement. Notice that, as a baseline approach, OPF takes the RP-based images in their raw format as input. The images generated by the recurrence plot approach have resolutions of 64×64 and 128×128, which turns out to produce feature vectors with $64 \times 64 = 4096$ and $128 \times 128 = 16{,}384$ dimensions to be handled by OPF, respectively. Although OPF is not based on distances explicitly, it uses them to compute the power of connectedness among samples to rule its competition process. Therefore, it is expected to obtain worst results when the number of dimensions increases, as one can observe in Table 2.

Table 2
Mean overall accuracies concerning the Meander dataset.

Proposed approach				
	50%/50% (train/test)		75%/25% (train/test)	
	64×64	128×128	64×64	128×128
ImageNet	**83.54%**	**<u>83.93%</u>**	**88.05%**	**<u>87.66%</u>**
CIFAR-10	78.90%	81.30%	**<u>86.36%</u>**	84.87%
LeNet	**84.03%**	23.73%	85.06%	25.45%
OPF	66.61%	63.54%	70.74%	63.90%
Approach proposed by Pereira et al. [29]				
	50%/50% (train/test)		75%/25% (train/test)	
	64×64	128×128	64×64	128×128
ImageNet	**<u>86.14%</u>**	**<u>84.74%</u>**	**85.00%**	**<u>87.14%</u>**
CIFAR-10	56.59%	50.00%	68.83%	64.22%
LeNet	25.45%	40.00%	43.64%	36.36%
OPF	79.87%	76.62%	**84.42%**	81.82%

Table 3 presents the mean sensitivity and specificity values regarding the Meander dataset. One can observe that CIFAR-10 and LeNet figured among the best accuracies, achieving average

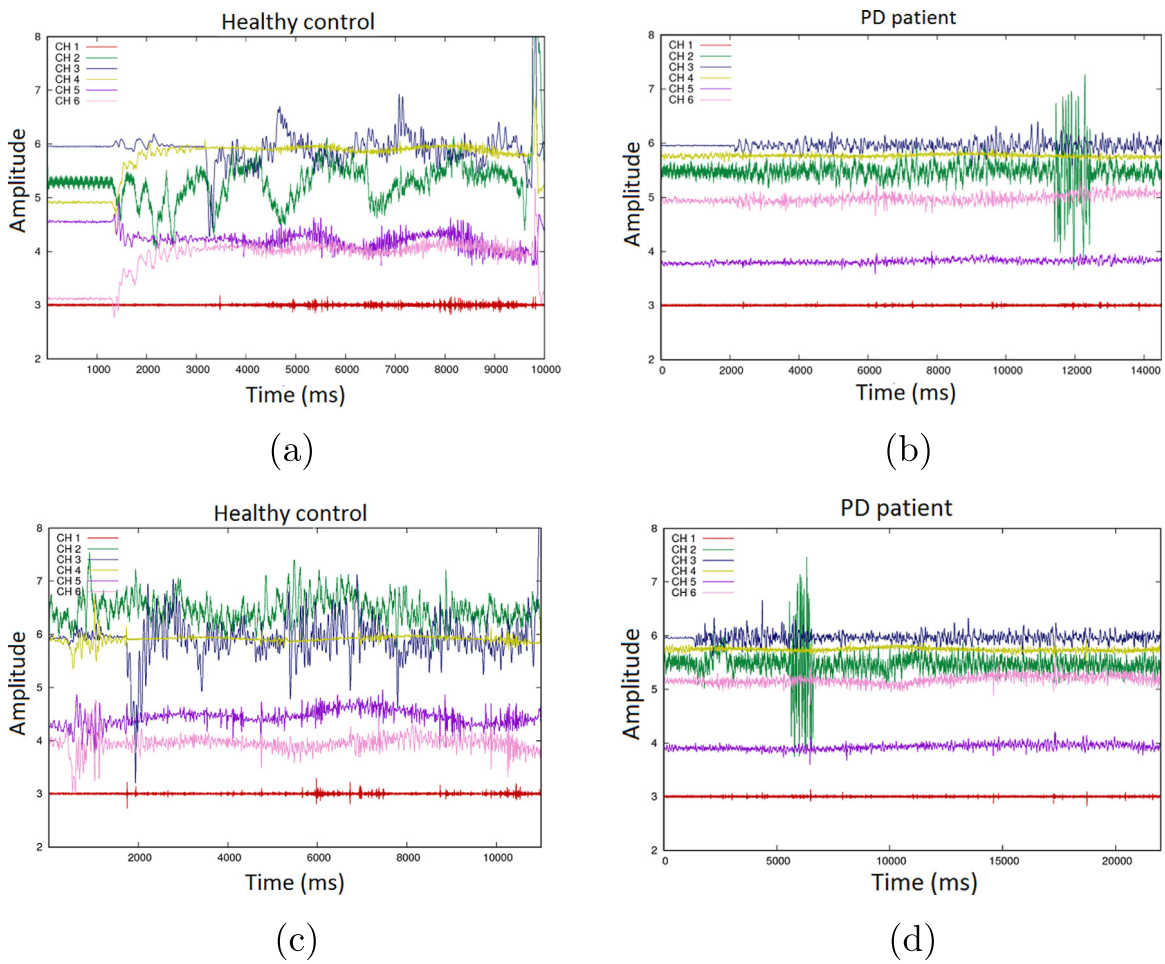

Fig. 4. Signals acquired during the handwriting exam: (a) corresponding to Fig. 3a, (b) corresponding to Fig. 3b, (c) corresponding to Fig. 3c, and (d) corresponding to Fig. 3d. The channels represent signals of the microphone (CH 1), fingergrip (CH 2), axial pressure of ink refill (CH 3), and tilt and acceleration in "X" (CH 4), "Y" (CH 5), and "Z" (CH 6) directions, respectively.

gains of 48.73% and 35.30% for the HC class with respect to the approach proposed by Pereira et al. [29], respectively. The exceptions regarding the mentioned class were ImageNet and OPF. The highest accuracies are 96.88% for the PD class and 73.57% concerning the HC class, both obtained by ImageNet. The OPF also showed competitive results in the PD class with accuracies of over 90%.

Tables 4 and 5 provide more details on the performance of each technique. The first one presents the values for precision, recall, and F1-measure with the best ones shown bolded. ImageNet figured among the highest values of F1-measure whereas, OPF achieved the best precision values. One can notice the increase in resolution has a significant impact in LeNet's performance with a significative decrease of the values for the computed metrics. Table 5 presents the average training times of each technique. OPF took longer than other techniques because of the input dimension, which is the total number of pixels of the RP image.

3.2. Spiral dataset

In this section, we discuss the results concerning the Spiral dataset, as presented in Table 6. Such data posed a more significant challenge than Meander dataset in our previous work [29]. However, the CNN-based architectures using our proposed approach showed improvements in nearly all scenarios, thus reducing the accuracy gap between Meander and Spiral datasets. Once again, the most significant improvements were registered by the LeNet architecture using images of resolution 64×64 (Table 6) with gains of 26.88% for a training set of size 50% and 41.55% for a training set of size 75%. However, the increasing of the image resolution led LeNet to poor performances on both approaches, being considerably higher in the proposed approach.

Table 7 presents the mean sensitivity and specificity values concerning the Spiral dataset. An improvement can be observed in the Spiral dataset with a significant increase of accuracy to all CNN-based techniques in almost all situations in the HC class. The main improvement (i.e., gain) was achieved by CIFAR-10 (average of 39.58%) followed by LeNet (32.20%). Also, the highest accuracies in both classes were obtained by ImageNet, with 72.86% and 97.86% of recognition rates concerning HC and PD, respectively.

As well as in the Meander dataset, ImageNet showed the highest F1-measure values followed by CIFAR-10 (i.e., bolded values in Table 8). Also, images of a higher resolution (128×128) presented a problem for the LeNet architecture whose accuracy performance significantly decayed. Table 9 presents the average training time of each technique. The behavior of each technique is similar to the one in the Meander dataset.

3.3. Discussion

In general, the experimental results showed LeNet to be very sensitive to the increase of image resolution mainly because of its shallower architecture. Differently, CIFAR-10 and ImageNet achieved a slight improvement in the same situation. RP-based

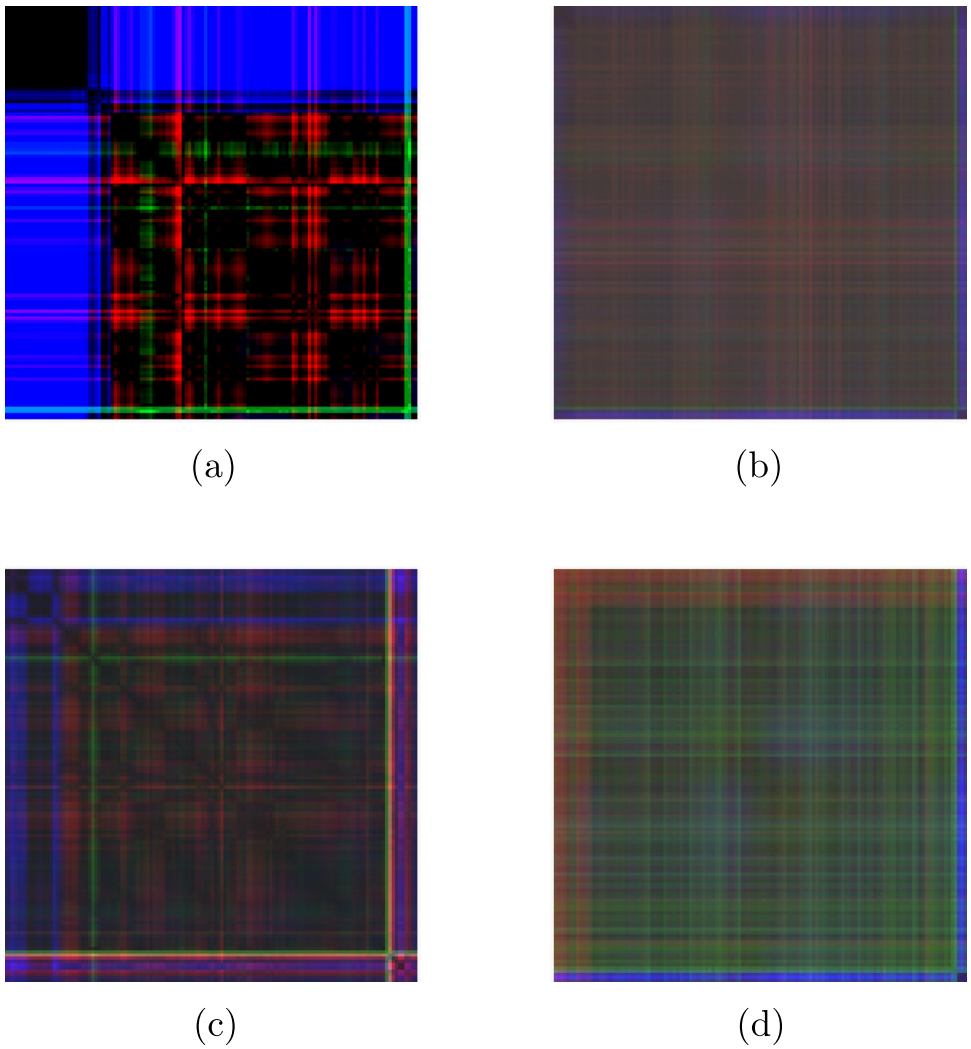

Fig. 5. RP-based samples: (a) HC and (b) PD patient concerning Meander, (c) HC and (d) PD patient concerning Spiral.

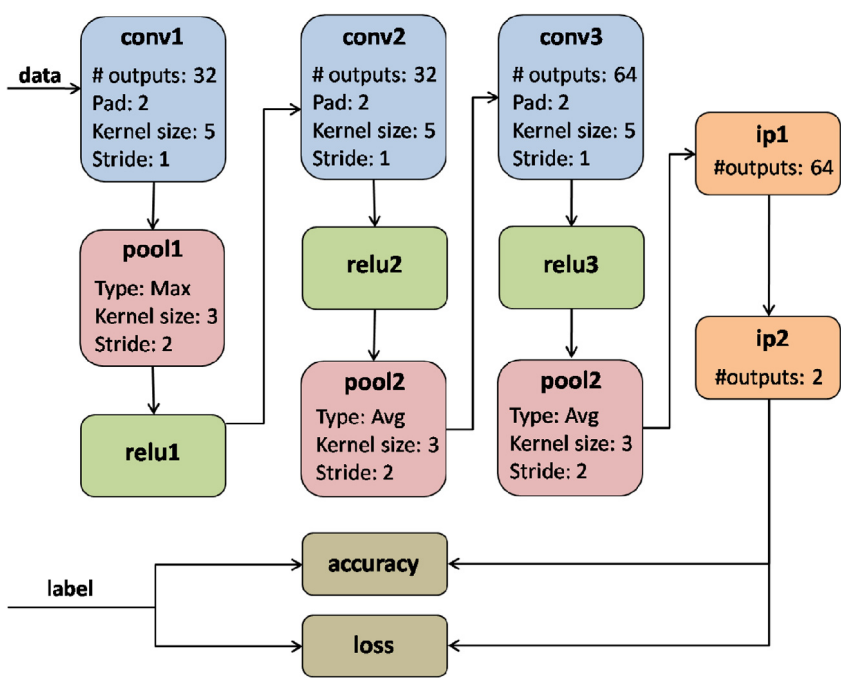

Fig. 6. CIFAR10_quick architecture.

Table 3
Mean sensitivity and specificity values concerning the Meander dataset.

Proposed approach								
	50%/50% (train/test)				75%/25% (train/test)			
	64 × 64		128 × 128		64 × 64		128 × 128	
	HC	PD	HC	PD	HC	PD	HC	PD
ImageNet	58.93%	**92.77%**	62.98%	91.79%	64.52%	**96.88%**	**73.57%**	**92.95%**
CIFAR-10	61.19%	85.54%	**66.55%**	86.83%	**71.67%**	91.88%	65.48%	**92.14%**
LeNet	**68.93%**	89.69%	09.64%	29.02%	62.62%	93.48%	00.00%	35.00%
OPF	40.24%	**92.99%**	33.33%	**93.75%**	46.67%	94.82%	34.76%	**93.04%**
Approach proposed by Pereira et al. [29]								
	50%/50% (train/test)				75%/25% (train/test)			
	64 × 64		128 × 128		64 × 64		128 × 128	
	HC	PD	HC	PD	HC	PD	HC	PD
ImageNet	**74.29%**	**90.58%**	**76.31%**	**87.90%**	**72.86%**	89.55%	**76.19%**	91.25%
CIFAR-10	15.71%	71.92%	10.00%	65.00%	33.33%	82.14%	10.95%	84.20%
LeNet	00.00%	35.00%	00.00%	55.00%	00.00%	60.00%	00.00%	50.00%
OPF	61.91%	86.61%	52.38%	85.71%	61.91%	**92.86%**	52.38%	**92.86%**

Table 4
Precision, recall and F1-measure concerning the Meander dataset.

Proposed approach												
	50%/50% (train/test)						75%/25% (train/test)					
	64 × 64			128 × 128			64 × 64			128 × 128		
	Precision	Recall	F1-measure	Precision	Recall	F1-measure	Precision	Recall	F1-measure	Precision	Recall	F1-measure
ImageNet	0.928	0.858	**0.891**	0.918	0.869	**0.893**	**0.969**	0.879	**0.922**	**0.929**	**0.904**	**0.916**
CIFAR-10	0.855	0.855	0.855	0.868	**0.874**	0.871	0.919	**0.896**	0.907	0.921	0.877	0.899
LeNet	0.897	**0.885**	**0.891**	0.290	0.461	0.356	0.935	0.870	0.901	0.350	0.483	0.406
OPF	**0.955**	0.793	0.866	**0.964**	0.806	0.878	0.946	0.841	0.891	**0.929**	0.825	0.874
Approach proposed by Pereira et al. [29]												
	50%/50% (train/test)						75%/25% (train/test)					
	64 × 64			128 × 128			64 × 64			128 × 128		
	Precision	Recall	F1-measure	Precision	Recall	F1-measure	Precision	Recall	F1-measure	Precision	Recall	F1-measure
ImageNet	**0.906**	**0.904**	**0.905**	**0.879**	**0.908**	**0.893**	0.896	**0.898**	0.897	**0.912**	**0.911**	**0.912**
CIFAR-10	0.719	0.695	0.707	0.650	0.658	0.654	0.821	0.767	0.793	0.842	0.716	0.774
LeNet	0.350	0.483	0.406	0.550	0.595	0.571	0.600	0.615	0.608	0.500	0.571	0.533
OPF	0.875	0.831	0.852	0.830	0.845	0.838	**0.911**	0.895	**0.903**	0.857	0.889	0.873

Table 5
Mean training times concerning the Meander dataset [seconds].

	50%/50% (train/test)		75%/25% (train/test)	
	64 × 64	128 × 128	64 × 64	128 × 128
ImageNet	0.0046	0.0047	0.0107	0.0094
CIFAR-10	0.0041	0.0043	0.0093	0.0091
LeNet	0.0041	0.0042	0.0089	0.0087
OPF	0.1238	0.4811	0.2864	1.1266

Table 6
Mean overall accuracies concerning the Spiral dataset.

Proposed approach				
	50%/50% (train/test)		75%/25% (train/test)	
	64 × 64	128 × 128	64 × 64	128 × 128
ImageNet	**83.70%**	**82.05%**	**86.69%**	**87.73%**
CIFAR-10	80.91%	**81.33%**	**86.43%**	**87.60%**
LeNet	81.43%	19.09%	85.19%	15.39%
OPF	64.90%	62.40%	67.49%	64.32%
Approach proposed by Pereira et al. [29]				
	50%/50% (train/test)		75%/25% (train/test)	
	64 × 64	128 × 128	64 × 64	128 × 128
ImageNet	78.41%	77.69%	**80.19%**	**77.53%**
CIFAR-10	75.58%	73.38%	**78.31%**	70.78%
LeNet	54.55%	40.00%	43.64%	40.00%
OPF	**83.77%**	**80.52%**	**79.22%**	**77.92%**

representations contain more information, and high-dimensional inputs require deeper architectures to better learn features.

However, concerning the Meander dataset, both CIFAR-10 and ImageNet models benefited from higher-resolution images, but just in case of smaller training sets (i.e., with 50% of the whole dataset size). Although larger training sets allowed better recognition rates, the neural networks did not outperform the results using lower-resolution images. In this case, larger training sets (i.e., 75%) seem not to be enough to alleviate the problem related to higher-resolution images since we have more information to deal with.

However, the behavior mentioned above cannot be observed for the Spiral dataset, but still, the results are quite close to a precise judgment. The best results considering spiral drawings are slightly worse than the ones obtained with meanders, which means spirals may be easier to draw than meanders. The latter figure straight lines that are not straightforward to draw, as well as the corners also present a challenge since one needs to change the direction of the handwriting abruptly.

An in-depth analysis was also provided by examining the sensitivity and specificity values, as presented in Tables 3 and 7. The accuracy of HC and PD classes were enhanced by the proposed approach in most cases. The most positives variations are noticed for CIFAR-10 and LeNet architectures, especially for the HC class in both datasets. The problem concerns the patients at the beginning of their appointments since they still not present the first symptoms of Parkinson's disease. Therefore, their drawings are pretty much similar to those from the control group.

Table 7
Mean sensitivity and specificity values concerning the Spiral dataset.

Proposed approach								
	50%/50% (train/test)				75%/25% (train/test)			
	64 × 64		128 × 128		64 × 64		128 × 128	
	HC	PD	HC	PD	HC	PD	HC	PD
ImageNet	55.24%	**94.38%**	**60.36%**	90.18%	56.90%	**97.86%**	**72.86%**	93.30%
CIFAR-10	**65.48%**	86.70%	**58.21%**	90.00%	**70.71%**	92.32%	**72.14%**	93.39%
LeNet	54.17%	91.65%	04.88%	24.42%	66.43%	92.23%	03.33%	19.91%
OPF	35.24%	**94.55%**	29.17%	**95.62%**	40.24%	94.73%	33.10%	**95.54%**
Approach proposed by Pereira et al. [29]								
	50%/50% (train/test)				75%/25% (train/test)			
	64 × 64		128 × 128		64 × 64		128 × 128	
	HC	PD	HC	PD	HC	PD	HC	PD
ImageNet	58.10%	86.03%	56.19%	85.76%	59.52%	**87.95%**	55.48%	85.80%
CIFAR-10	29.76%	**92.77%**	06.31%	**98.53%**	57.38%	86.16%	14.76%	**91.79%**
LeNet	00.00%	75.00%	00.00%	55.00%	00.00%	60.00%	00.00%	55.00%
OPF	**66.67%**	90.18%	**61.91%**	87.50%	**71.43%**	87.50%	**66.67%**	82.14%

Table 8
Precision, recall and F1-measure concerning the Spiral dataset.

Proposed approach												
	50%/50% (train/test)						75%/25% (train/test)					
	64 × 64			128 × 128			64 × 64			128 × 128		
	Precision	Recall	F1-measure	Precision	Recall	F1-measure	Precision	Recall	F1-measure	Precision	Recall	F1-measure
ImageNet	0.944	0.849	**0.894**	0.902	**0.858**	**0.880**	**0.979**	0.858	**0.914**	0.933	**0.902**	**0.917**
CIFAR-10	0.867	**0.870**	0.869	0.900	0.852	0.875	0.923	**0.894**	0.908	**0.934**	0.899	0.916
LeNet	0.917	0.842	0.878	0.244	0.406	0.305	0.922	0.880	0.901	0.199	0.355	0.255
OPF	**0.946**	0.822	0.880	**0.955**	0.781	0.859	0.946	0.815	0.876	0.929	0.825	0.874
Approach proposed by Pereira et al. [29]												
	50%/50% (train/test)						75%/25% (train/test)					
	64 × 64			128 × 128			64 × 64			128 × 128		
	Precision	Recall	F1-measure	Precision	Recall	F1-measure	Precision	Recall	F1-measure	Precision	Recall	F1-measure
ImageNet	0.860	0.846	0.853	0.858	**0.839**	**0.848**	**0.879**	0.853	0.866	0.858	0.837	0.847
CIFAR-10	**0.928**	0.779	0.847	**0.985**	0.737	0.843	0.862	0.844	0.852	0.918	0.742	0.820
LeNet	0.750	0.667	0.706	0.550	0.595	0.571	0.600	0.615	0.608	0.550	0.595	0.571
OPF	0.893	**0.847**	**0.870**	0.839	0.825	0.832	0.875	**0.875**	**0.875**	**0.929**	**0.852**	**0.889**

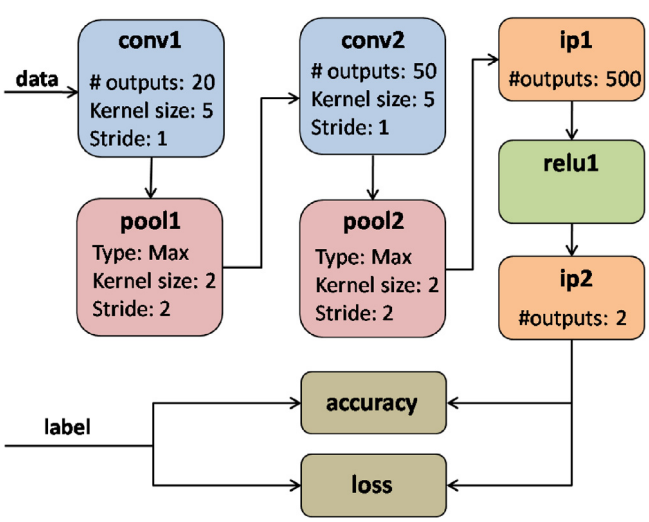

Fig. 7. LeNet architecture.

Table 9
Mean training times concerning the Spiral dataset [seconds].

	50%/50% (train/test)		75%/25% (train/test)	
	64 × 64	128 × 128	64 × 64	128 × 128
ImageNet	0.0046	0.0044	0.0100	0.0104
CIFAR-10	0.0043	0.0041	0.0093	0.0094
LeNet	0.0042	0.0041	0.0089	0.0090
OPF	0.1229	0.4818	0.2758	1.0795

4. Conclusions and future works

This work introduced the recurrence plot for modeling signals to automatic Parkinson's disease identification. The signals are referred as patient's handwriting dynamics recorded during an exam, which are further used as input to a CNN that identifies whether an individual has or not PD. The main contribution of this work is the improvement of the precision of PD identification using deep learning-based approaches.

Experimental results showed satisfactory and significant improvement of the accuracy rates considering all deep-based architectures in most scenarios, especially in the HC class that showed to be a very challenging one due to the existence of samples that are similar to the ones of PD patients. It was also observed that higher-resolution images do not guarantee better accuracies. Moreover, the additional information provided by the recurrence plot supported the improvement of accuracies.

Since our main concern relies on the PD identification, the proposed approach achieved a satisfactory improvement in most of the situations, reaching accuracy rates of over 90%. Therefore, RP-based images provided more useful information that enabled the machine learning methods to identify PD samples better.

Future works include the study and development of web and mobile platform capable of managing and monitoring patients' data using the Internet of Health Things (IoHT) [36].

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